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Predicting Airline Satisfaction

Kaggle Playground Series S6E10 · binary classification · metric: ROC AUC

Data & EDA
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Preprocessing & validation
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Modelling & submissions
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Task and data

Target

Is the passenger **satisfied**? 44.4% yes, 55.6% no — mildly imbalanced, no resampling needed.

Metric: ROC AUC

The chance a random satisfied passenger is scored above a random unsatisfied one. Rewards correct **ranking**; 0.5 = guessing.

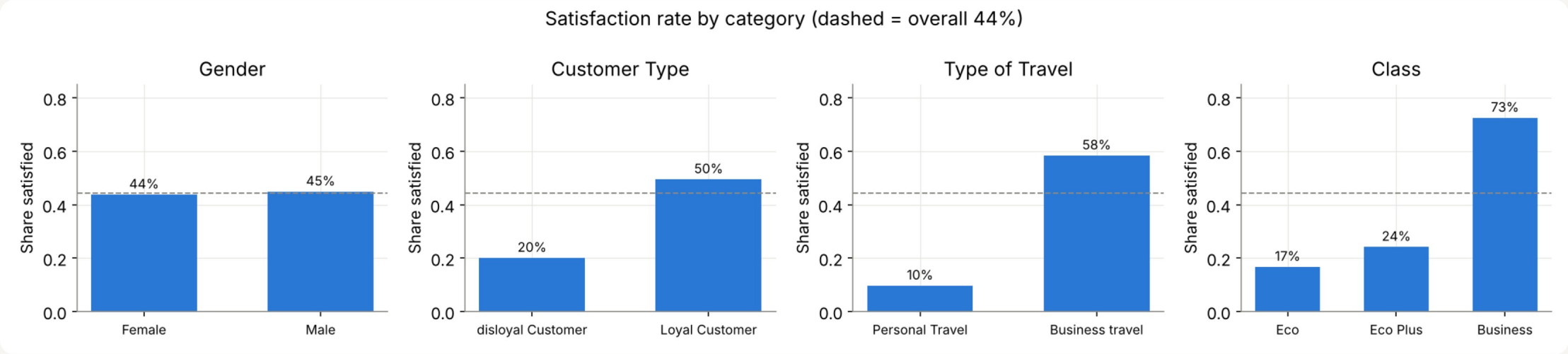
Data

700k train, 300k test rows. 22 features: 4 categorical, 4 numeric, 13 service ratings (0–5). Synthetic, from a real survey.

Data problem	How big	What we do
Missing arrival delay	292 train, 130 test	Fill with departure delay (r = 0.84) + flag
Rating 0 = "not applicable"	up to 3.7% per column	Keep as own level, count per passenger
Skewed delays	91% zero, max 490 min	Keep; log1p for the linear model

Source: Kaggle competition data · notebook 01

Key EDA findings



Trip context dominates

Business class 73% satisfied vs Eco 17%. Business travel 58% vs personal 10%.

Online boarding is #1

Strongest rating (r = 0.56). Satisfaction jumps at 4–5 — not linear.

0 is not a low score

Wifi rated 0 (N/A): 89% satisfied. Kept as its own level.

Dashed line = overall 44% satisfied · Gender and delays barely matter · notebook 01

Validation and current scores

Validation

5-fold stratified CV, same folds for every model, out-of-fold ROC AUC.

No leakage

Scaling and one-hot sit in an sklearn Pipeline, refit on each training fold.

Encoding

Binary features → 0/1. Class → one-hot (linear) or ordered 0–2 (trees).

Model	CV AUC
Constant (trivial baseline)	0.5000
Rule: online boarding rating	0.8404
Rule: business class + boarding	0.8802
Logistic regression	0.9368
LightGBM (untuned)	0.9587
Kaggle public leaderboard (LightGBM)	[0.____]

Fold std ≤ 0.0011 for every model · notebooks 02 and 03

Plan to the defense

HARISSON TENU
DATA & FEATURES

Feature engineering

- Mean, min and count-of-5 ratings
- Rating × class / travel interactions
- Drop weak features (gender, delays)
- Measure each change on CV

EMMANUEL GYAMFI KWARTENG
VALIDATION & MODELS

More model types

- Random forest
- XGBoost and CatBoost
- One comparison table, same folds
- Leakage checks on new features

ENO DANIEL SIMEON
TUNING & SUBMISSION

Tune and choose

- Random search on LightGBM
- Blend of the best models
- Compare CV with leaderboard
- Final model choice and why

Together: the defense slides, each member presenting their own part (Dec 4 / 11).